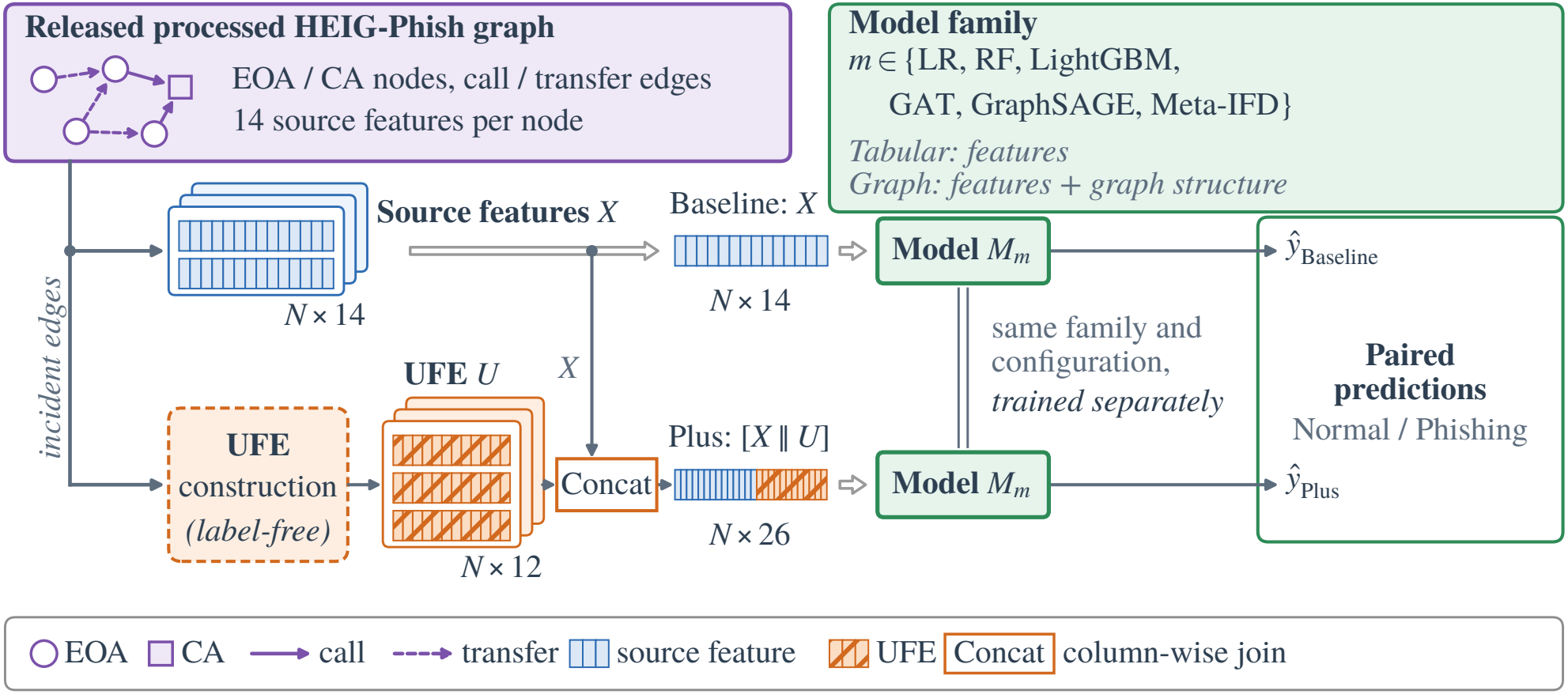
Released processed HEIG-Phish graph
EOA / CA nodes, call / transfer edges
14 source features per node
Model family
$m \in$ {LR, RF, LightGBM,
GAT, GraphSAGE, Meta-IFD}
Tabular: features
Graph: features + graph structure
incident edges
Source features X
$N \times 14$
Baseline: X
$N \times 14$
Model M_m
$\hat{y}_{\text{Baseline}}$
same family and
configuration,
trained separately
Paired
predictions
Normal / Phishing
UFE construction
(label-free)
UFE U
$N \times 12$
X
Concat
Plus: $[X \,\|\, U]$
$N \times 26$
Model M_m
$\hat{y}_{\text{Plus}}$
EOA
CA
call
transfer
source feature
UFE
Concat
column-wise join